

TM Forum Component

Product Inventory

TMFC005

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Notice

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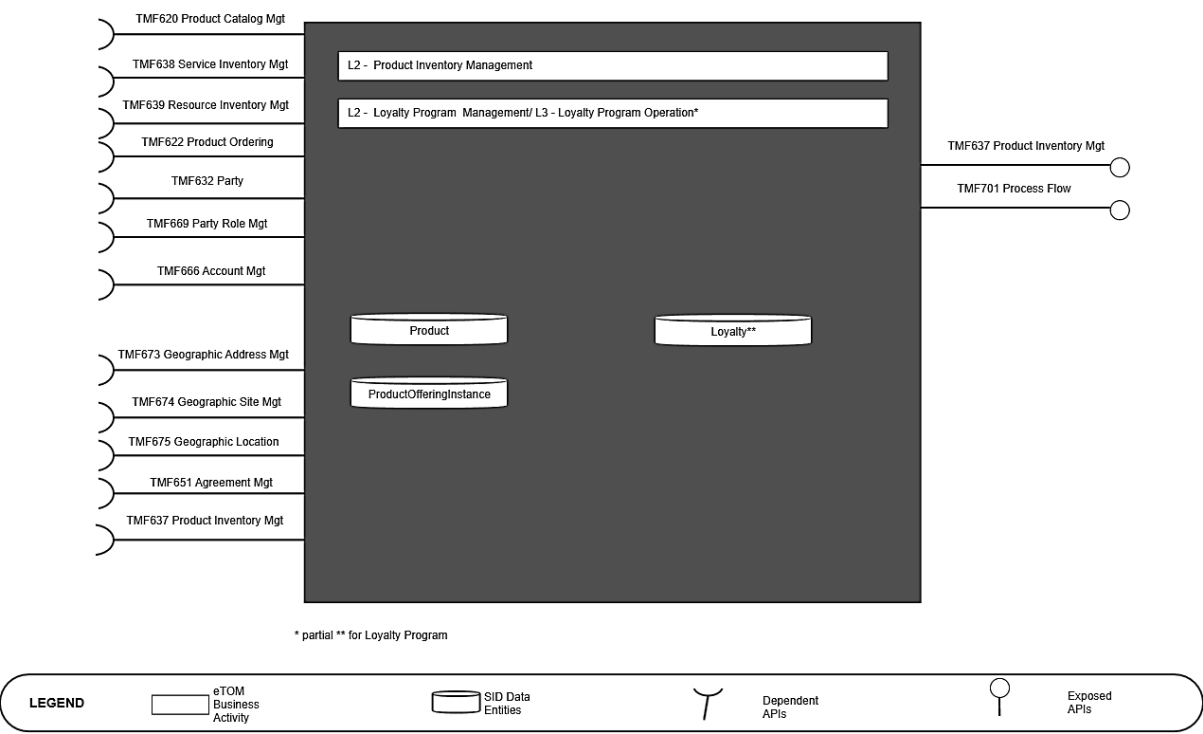
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1. Overview

Component Name	ID	Description	ODA Function Block
Product Inventory	TMFC005	The Product Inventory component is responsible for storage and exposure of products that are assigned to and used by Parties. This component has functionality that enables creation of inventory items, inventory organization, inventory search or filter, inventory monitoring and tracking, inventory control and inventory auditing. The minimum check to be performed upon inventory item creation or update is for global consistency with related Product Catalog information.	Core Commerce Management

TMFC005 : Product Inventory



2. eTOM Processes, SID

Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

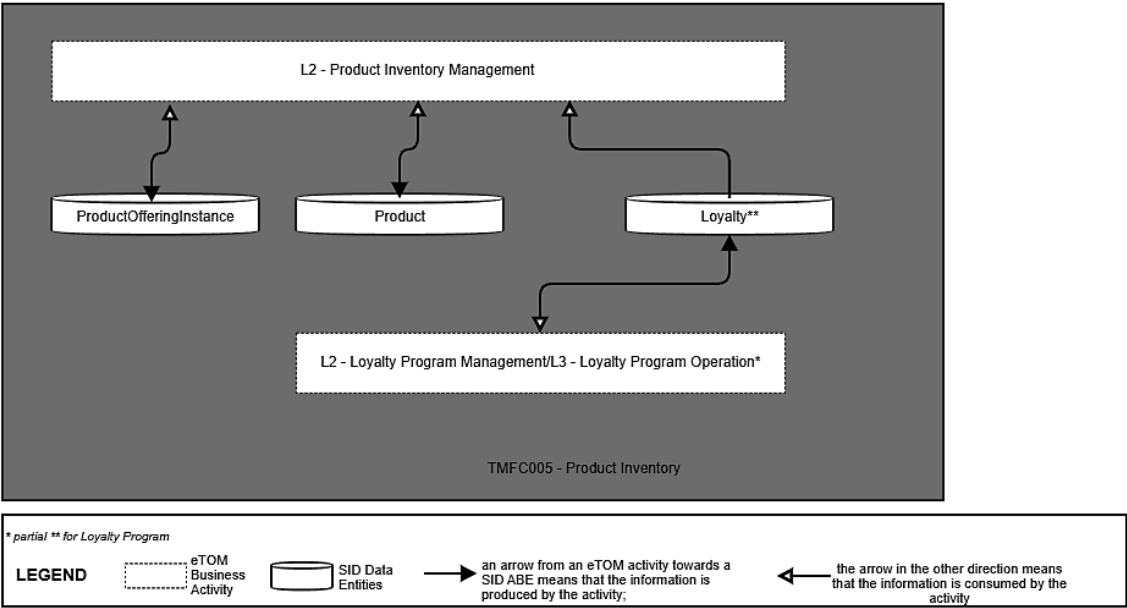
Identifier	Level	Business Activity Name	Description
1.2.11	L2	Product Inventory Management	Product Inventory Management is responsible to establish, manage and administer the enterprise's product inventory, as embodied in the Product Inventory repository, and monitor and report on the usage and access to the product inventory, and the quality of the information maintained in it.
1.1.19	L2	Loyalty Program Management	Define all aspects of a loyalty program, such as requirements and objectives of a loyalty program, determine the benefits to participants. Develop a program, prototype it, test it, rollout/launch it, amend and evaluate it, and terminate it when it is no longer viable for an enterprise. Manage all operational aspects of running a loyalty program. Enable parties to become a members of a program, earn currency and rewards, and redeem currency. Manage a loyalty program account, leave a program, and provide operational reports.
1.1.19.2	L3	Loyalty Program Operation	Manage all operational aspects of running a loyalty program. Enable parties to become a members of a program, earn currency and rewards, and redeem currency. Manage a loyalty program account, leave a program, and provide operational reports.
1.1.19.2.5	L4	Manage Loyalty Program Account	Update a loyalty program account and make changes to loyalty program participant information. Expire, reinstate, transfer in/out, adjust, a loyalty participant's account currency. Prepare and send a loyalty program communication to a participant or for internal use by an enterprise.
1.1.19.2.7	L4	Provide Loyalty Program Operation Report	Generate a loyalty program operation report, such as various loyalty program status reports, trend analysis, and reports that identify suspected abuse of a loyalty program.

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)
Product	
ProductOfferingInstance	
Loyalty	Loyalty Program

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
180	Assigned Products Maintenance	Assigned Products Maintenance permits defining and update: - product characteristics - links with the related service or resource (handsets, SIM cards, ...) needed to deliver the product, ...	ProductRepository Management	ProductInventory Repository Management
197	Customer Product Storage	Customer Product Storage provides the functionality necessary to store and make available the Products. services presently being used by the customer. This function allows: • to instantiate or update offers and products ordered by the customer, whatever their type (network product, bundle, device, ...) or their marketing mode (rented, sold, ...), with their configuration, their tariffs and discounts, and their status (initialized with a creation order) • to update Products status • to search and read Offer and Product installed base (subscribed offers, configuration of installed products, installed tariffs and discount, statuses, ...).	ProductRepository Management	ProductInventory Repository Management
198	Customer Loyalty Score Balance Management	Customer Loyalty Score Balance Management function calculates the score according to accumulation/decrease rules. When a customer subscribes to the loyalty program with more than one SIM or other 'traffic objects', the Score Management accumulates the points into a single balance. The loyalty score could decrease for one of	ProductRepository Management	Loyalty Account Management

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		the following events: prize purchase, points expiry or points deletion by Call Centre. The functionality may also include the visualization of Score details (date, description event type, points, final score) via different contact Channels (e.g. via Web, IVR, Call Centre).		
237	Customer Loyalty Communication	Customer Loyalty Communication function sends information related to Loyalty Programs (Point Balance, Prize Request status, renewed Loyalty Code) to external components in push and pull modes.	ProductRepository Management	Loyalty Account Management
361	Contract Implementation P Product Agreement Implementation	Contract Implementation function provide functionality pertaining to the implementation of the contract across fulfillment, assurance, and billing. Product Agreement Implementation function provides functionality pertaining to the implementation of the Product Agreement (a.k.a. contract) across fulfillment, assurance, and billing according to Product Agreement Specification. A Product Agreement represents the approval by the Customer and the Vendor of all term or conditions of a ProductOffering.	ProductRepository Management	ProductInventory Agreement Management
362	Contract Searching	Contract Searching function provides the ability to search for customer contracts based on meta-data and to search text strings within contracts and view customer's existing and previous contracts;	ProductRepository Management	ProductInventory Management
363	Contract Storage Product Agreement Storage	Contract Storage provides the central repository for contract storage as well as the associated contract meta-data.	ProductRepository Management	ProductInventory Agreement Management

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		This data can be mined for Campaigns and Lead Generation. Product Agreement Storage provides functionality necessary to store and make available the Product Agreements. This function allows: • to instantiate or update Product Agreement approved by the customer with their party involved, their configuration, their approvals and their status, • to update Product Agreements status, • to search and read Product Agreements.		
1201	Product Configuration Check	Product Configuration Check Function Checks for each Product submitted if all Product configuration rules have been respected such as prerequisite, incompatibility rules between Product Specifications or mandatory characteristics.	Product Management	Product Repository Management

3. TM Forum Open APIs & Events

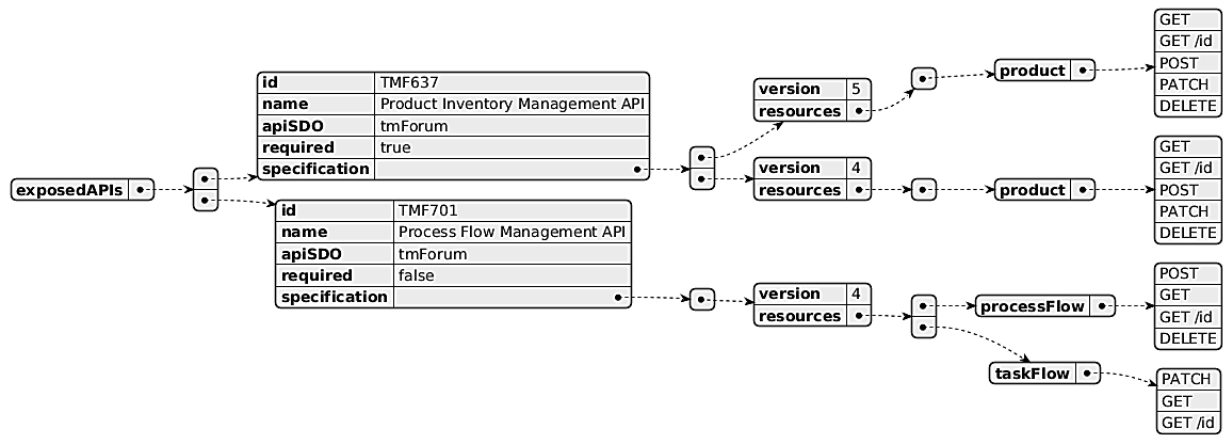
The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.



3.1.Exposed APIs

The following diagram illustrates API/Resource/Operation:

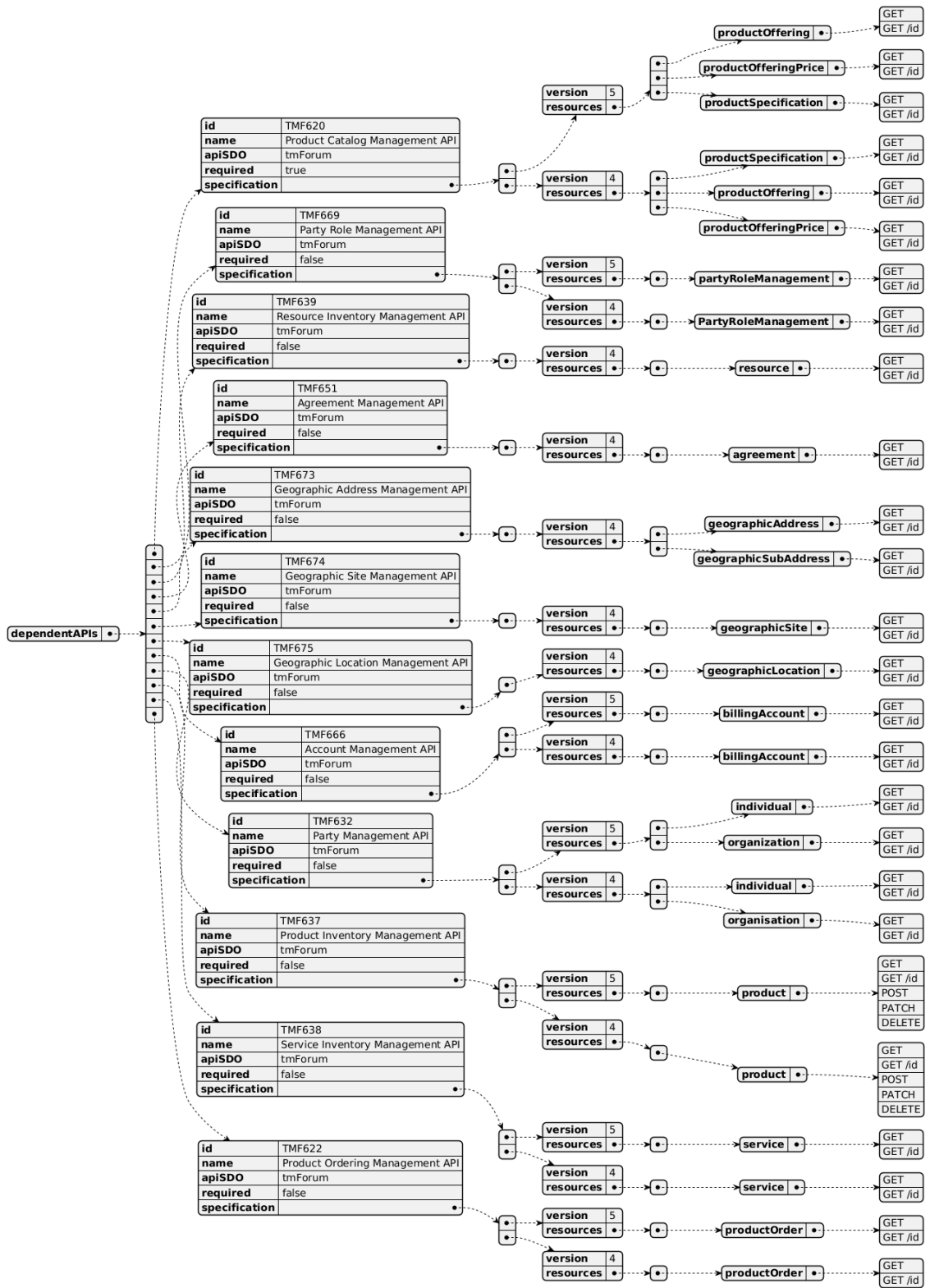


API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF637	Product Inventory Management	Mandatory	4	Product	GET GET /ID POST PATCH DELETE
		Mandatory	5	Product	GET GET /ID POST PATCH DELETE
TMF701	Process Flow Management	Optional	4	taskFlow	GET GET /ID PATCH
				processFlow	GET GET /ID POST DELETE



3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation:

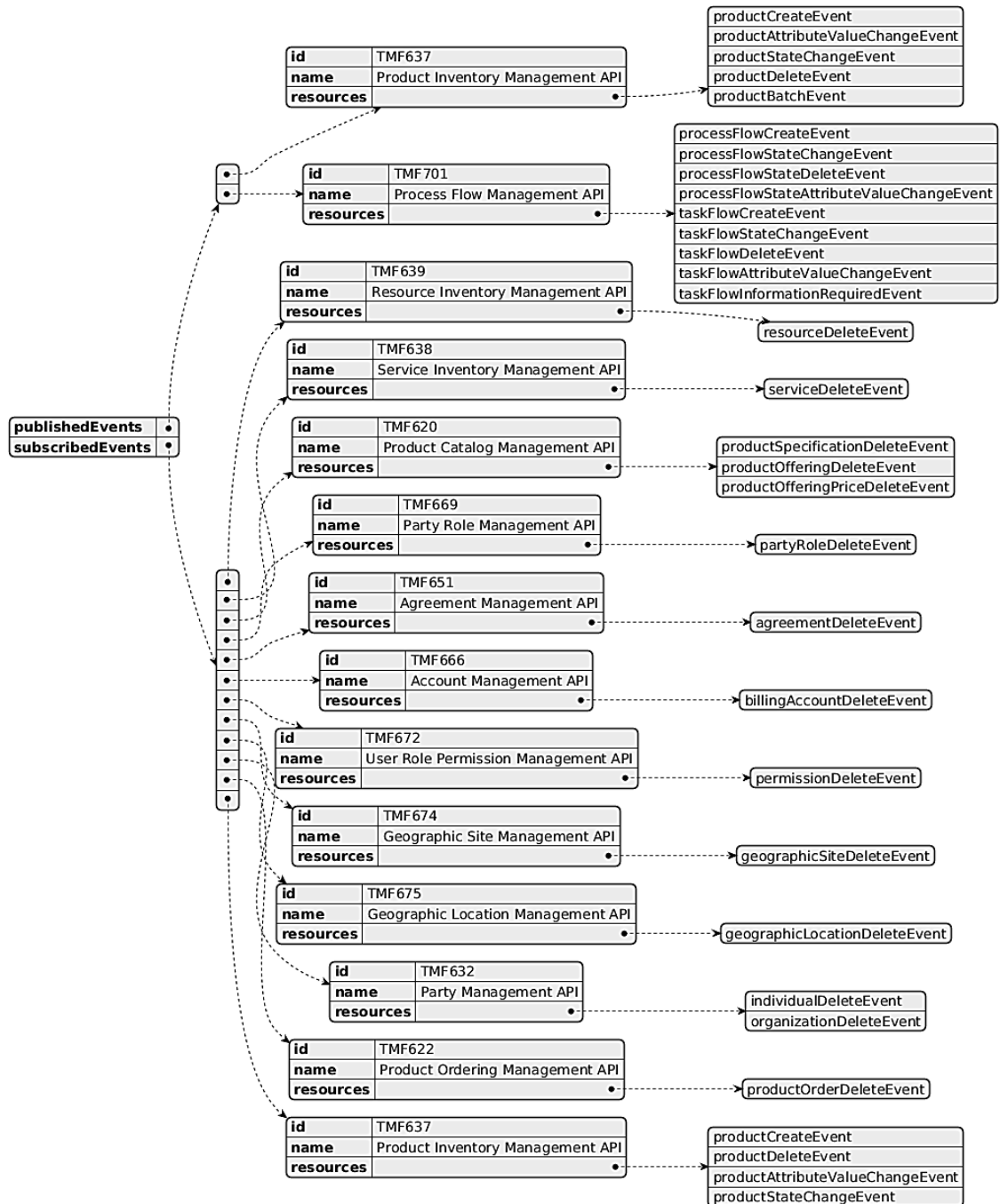


API ID	API Name	API Version	Mandatory / Optional	Resource	Operation
TMF666	Account Management	4	Optional	billingAccount	Get Get/id
TMF669	Party Role Management	4	Optional	partyRole	Get Get/id
		5	Optional	partyRole	Get Get/id
TMF632	Party	4	Optional	individual	Get Get/id
				organization	Get Get/id
		5	Optional	individual	Get Get/id
				organization	Get Get/id
TMF672	User Roles And Permissions	4	Optional	permission	Get Get/id
TMF673	Geographic Address Management	4	Optional	geographicAddress	Get Get /id
				geographicSubAddress	Get Get /id
TMF674	Geographic Site Management	4	Optional	geographicSite	Get Get /id
TMF675	Geographic Location	4	Optional	geographicLocation	Get Get /id
TMF651	Agreement Management	4	Optional	agreement	Get Get /id
TMF639	Resource Inventory Management	4	Optional	resource	Get Get /id
TMF638	Service Inventory Management	4	Optional	service	Get Get /id
		5	Optional	service	Get Get /id
TMF620	Product Catalog Management	4	Mandatory	productSpecification	Get Get /id
				productOffering	Get Get /id
				productOfferingPrice	Get Get /id
		5	Mandatory	productSpecification	Get Get /id
				productOffering	Get Get /id
				productOfferingPrice	Get Get /id
TMF622	Product Ordering	4	Optional	productOrder	Get Get /id
		5	Optional	productOrder	Get Get /id

TMF637	Product Inventory	4	Optional	product	Get Get /id Post Patch Delete
		5	Optional	product	Get Get /id Post Patch Delete
TMF688	Event Management	4	Optional	event	Get Get /id

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Jira References

5.2.1. eTOM

- <https://projects.tmforum.org/jira/browse/ISA-413> L2 - Loyalty Program Management (1.1.19): review positioning of L2 (why in Market& Sales and not in Product) and detailed L3 and L4

5.2.2. SID

5.2.3. Functional Framework

- <https://projects.tmforum.org/jira/browse/ISA-415> Add a function to check data integrity at Product level, including mandatory relationships.
- <https://projects.tmforum.org/jira/browse/ISA-414> Clarify functions using "contract" in their name and/or definition as globally at product level (instanciation of Product Offering or Product Specification).
- <https://projects.tmforum.org/jira/browse/ISA-416> Harmonize as much as possible functions related to Product, Service and Resource Inventory

5.2.4. API

- TMF637- Product Inventory Management: list missing information and relationship as Loyalty Account
 - <https://projects.tmforum.org/jira/browse/AP-3844>
 - <https://projects.tmforum.org/jira/browse/AP-3139>
- TMF637 Product Inventory Management: list missing information related to SID ABE productOfferingInstance
 - <https://projects.tmforum.org/jira/browse/AP-7096>
- ProductPriceCharge should be managed by this component, including all specialization of ProductPriceCharge (oneTime, recussing, usage...)
 - See <https://projects.tmforum.org/jira/browse/ISA-1369>

5.3. Further resources

This component is involved in the following use cases described in [IG1228 How to use ODA - Using Open APIs to realize Use Cases](#):

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	31-May-2022	Kamal Maghsoudlou Sylvie Demarest	Final edits prior to publication
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.0.2	11-Jun-2024	Sylvie Demarest	Updated Component Template Aligned to Frameworks 23.5
1.0.3	28-Mar-2025	Gaetano Biancardi	Correct typo in Exposed API diagram
1.0.3	08-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025 Final edits prior to publication
1.0.4	30-Sep-2025	julien rouland	Update to reflect productOfferingInstance ABE as mentioned in ISA-414 - Clarify Product Inventory Mgt sub-domain vocabulary SME REVIEW
1.0.4	19-Nov-2025	Rosie Wilson	Final edits prior to publication
1.1.0		Hugo Vaughan (TM Forum)	Gen 5 API incorporation
1.1.0	25-Jun-2025	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	31-May-2022	Goutham Babu	Initial release
Production	22-Jul-2022	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	14-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.0.2
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	08-Jul-2025	Rosie Wilson	New release 1.0.3
Production	05-Sep-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	19-Nov-2025	Rosie Wilson	New release 1.0.4
Production			
Pre-production	01-Jun-2026	Sannah Sibaya	Final edits prior to publishing

6.2. Acknowledgements

This document was prepared by the members of the TM Forum ODA Components & Canvas team.

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